

## **Part A :**

**Question 1 :** Define the role of a **Privacy/Security Node** in an AI agent. Why is it essential before sending user input to an LLM?

**Answer 1: Role of a Privacy/Security Node :** A Privacy/Security Node is the part of an AI agent that checks the user's input before it goes to the LLM. Its job is to see if the input contains sensitive or private information—like passwords, phone numbers, ID numbers, or company secrets.

It is essential because it stops confidential or risky data from being sent to the LLM, which may store logs or pass data to external systems. This protects the user's privacy, follows company rules, and avoids legal problems.

**Question 2 :** Explain how a **Content Moderation Node** improves trust and safety in AI systems. Provide two real-life examples where moderation is critical.

**Answer 2. Content Moderation Node :** A Content Moderation Node checks both user input and AI output for harmful content. It ensures the conversation stays safe, respectful, and follows rules.

It improves trust and safety by:

- Blocking hate speech, bullying, or threats
- Preventing harmful advice (self-harm, illegal activities)
- Keeping the system suitable for general users

Real-life examples where moderation is critical:

1. Social media chats (Facebook comments, YouTube) : to stop hate speech and harassment.
2. Kids' learning apps : to make sure no adult or unsafe content is shown to children.

**Question 3 :** Compare and contrast pre-processing filters (before LLM input) vs. post-processing filters (after LLM output). Which is more effective for:

1. Preventing confidential data leakage
2. Preventing slang/adult content

**Answer 3. Pre-processing vs. Post-processing Filters :** Pre-processing filters work before the text reaches the LLM. They clean the user's input by removing private or sensitive information and blocking unsafe instructions.

Post-processing filters work after the LLM has generated a response. They check the output to make sure it does not contain harmful, adult, or toxic content.

For preventing confidential data leakage, the pre-processing filter is more effective. This is because it removes private details before the LLM ever sees them, so the model cannot store, repeat, or misuse that information.

For preventing slang or adult content, the post-processing filter is more effective. Even if the user input is safe, the LLM might still generate inappropriate words, so the output must be checked and cleaned before sending it back to the user.

**Question 4 :** What risks remain even if an agent has both a Privacy Node and a Moderation Node? Suggest mitigation strategies.

**Answer 4. What risks still remain And Mitigation Strategies:** Even with both Privacy and Moderation Nodes, some risks still exist:

### Remaining Risks

- The LLM may still misunderstand and generate incorrect information.
- Hidden sensitive data might get through if patterns are unusual.
- Users may try to bypass the filters intentionally (jailbreaks).
- The system may block harmless content accidentally (false positives).
- The AI might still leak indirect/derived information (inference attacks).

### Mitigation Strategies

- Add continuous monitoring and update filters regularly.
- Use human review for high-risk tasks.
- Keep audit logs to track suspicious behavior.
- Train the LLM with better safety data.
- Add rate limits to reduce repeated attack attempts.
- Use multi-layer security (encryption, access control, data minimization).